DESIGN THINKING FOR INNOVATION [IPDE211]

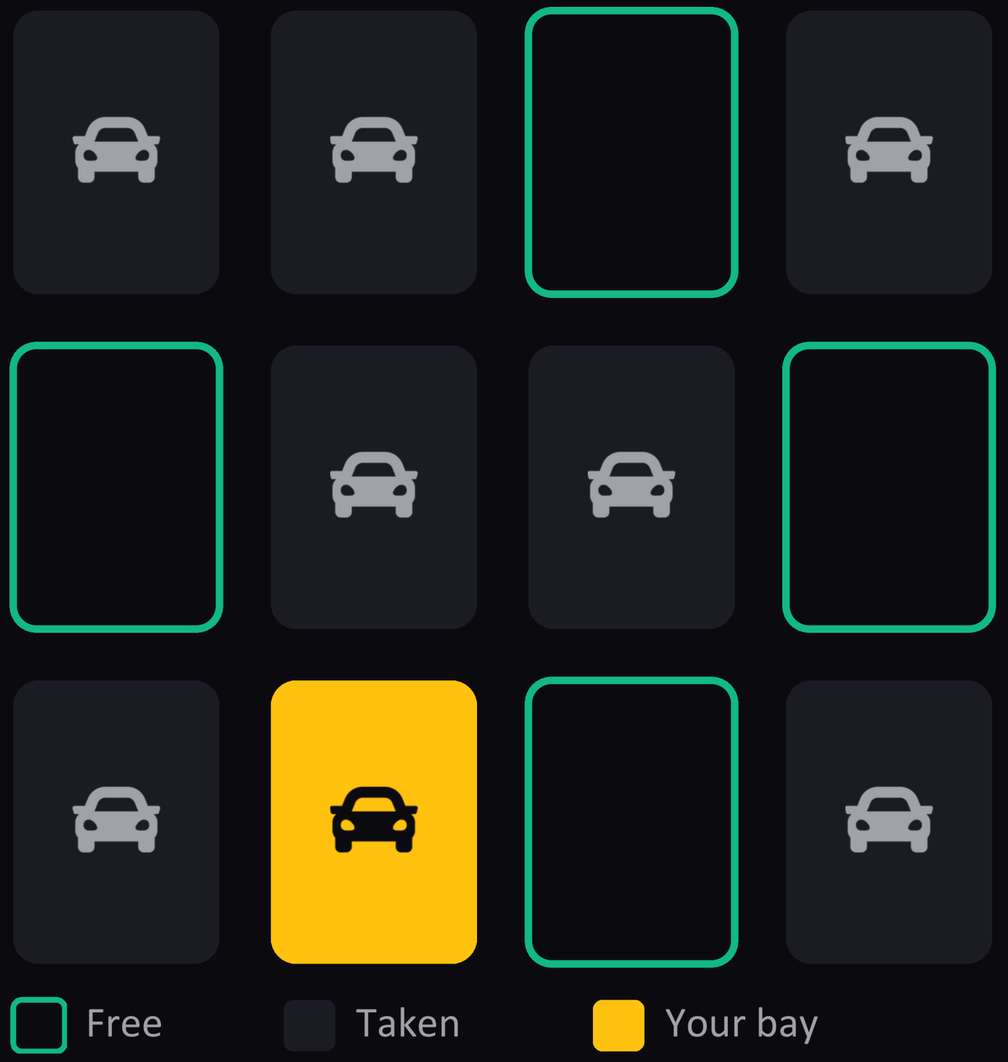
Application of Design Thinking in Solving Real-World Problems

ParkSure

Smart parking for Kolkata, without the wait

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PROBLEM IDENTIFICATION

The problem we chose

Finding parking in Kolkata is a search problem wrapped in a trust problem.

Unknown availability

Drivers cannot tell if a lot is free, so they circle and add to congestion.

Opaque pricing

A demand-based rate feels unfair unless the driver sees why it changed.

Two systems, one lot

Drivers and staff use different tools, so gate, cash and audit trail disagree.

PROBLEM DEFINITION

Our point of view

A driver near Park Street needs a trustworthy, live answer to "is there a bay, and what will it cost?", and the people running the lot need the same answer on their side of the barrier.

HOW MIGHT WE

give drivers and lot staff the same trustworthy answer before anyone reaches the barrier?

RESEARCH & TARGET GROUP

Who we designed for

PRIMARY USERS

The driver

PRESSURE

Circling in traffic, short on time

NEEDS

One clear answer on bay, price and gate

The attendant

PRESSURE

A queue building at the barrier

NEEDS

One unambiguous next action

ALSO SERVED

Operator

Needs a live lot map, alerts and cash reconciliation.

Administrator

Needs pricing rules, user roles and an audit trail.

IDEA GENERATION

Brainstorming ideas

Know before arriving	Trust the price	One answer, both sides
Live bay estimates with confidence scores	Demand-based pricing	Four role workspaces, one account
Camera estimates for open grounds	Price shown as a receipt	Shared audit trail
Map route to the lot	QR pass that opens the gate	Plate checks for attendants

Our filter: does the idea remove uncertainty, or show the reason behind a number?

PROPOSED SOLUTION

One account, four workspaces

Driver

Live bay estimates, a visible price breakdown, booking, QR pass and a GST receipt.

Attendant

Plate checks, manual entry and cash collection, one decision at a time.

Operator

Live lot map, alerts, an event log and end of shift cash reconciliation.

Administrator

Occupancy and pricing thresholds, user roles, audit trail and CSV exports.

PROPOSED SOLUTION: PRICING

A price you can read

Base rate set by the administrator	₹60 / hr
Occupancy lift 83% full against a 60% demand floor	x 1.20
Peak window weekday evening, 5 to 9 pm	+ 0.15
Multiplier 1.20 + 0.15	x 1.35
Live rate	**₹80 / hr**

Visible reasons

Drivers see the arithmetic behind a variable price.

Admin control

Base rate and thresholds are set by the lot administrator.

Built on trust

The same receipt logic is shown on every screen.

PROPOSED SOLUTION: BUILD & TEST

From sketch to prototype

1

Empathize

Two users, opposite pressures

2

Define

Point of view and a four role model

3

Ideate

Visible pricing, camera counts, audit trail

4

Prototype

Wireframes, then a working React app

5

Test

Contrast, link, mobile and smoke checks

BUILT WITH

React 19 | TypeScript | Tailwind CSS | Leaflet + OpenStreetMap | PWA + Android

EXPECTED IMPACT & BENEFITS

Who gains, and how

Drivers
Less circling and a price they can understand

Attendants
One clear decision at a time, even in a queue

Operators and administrators
One shared record of gate, cash and audit trail

The city
Fewer cars cruising for a bay on busy streets

BUILT FOR KOLKATA

4 locations: Park Street, Salt Lake, Howrah, Maidan

3 languages: English, Bengali, Hindi

WHAT'S NEXT

From prototype to pilot

1. Add a real backend and authentication
2. Connect live sensor and camera data
3. Pilot at one lot with real drivers and staff

Demo mode today

Occupancy, payments, refunds and gate openings are simulated. Data lives in the browser.

prateekcandwill.github.io/parksure-smart-parking-kolkata

github.com/Prateekcandwill/parksure-smart-parking-kolkata

Thank you